

Efficient Heuristics for Classical Simulation of Quantum Circuits Using ZX-Calculus

Candidate no. 1081502

Thesis word count: TODO

*Submitted in partial completion of the
Master of Science in Advanced Computer Science*

Trinity 2024

Abstract

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I hereby certify that this is entirely
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Acknowledgements

Personal

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List of Abbreviations

- 1-D, 2-D** . . . One- or two-dimensional, referring in this thesis to spatial dimensions in an image.
- Otter** One of the finest of water mammals.
- Hedgehog** . . . Quite a nice prickly friend.

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There is no one who loves pain itself, who seeks after it and wants to have it, simply because it is pain...

— Cicero's *de Finibus Bonorum et Malorum*

1

Introduction

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1.1 Motivation

Quantum computing, a field rooted in the principles of quantum mechanics, represents a revolutionary approach to computation. Unlike classical computing, which relies on binary operations performed on bits, quantum computing utilizes quantum bits, or qubits, capable of representing and processing information in ways that classical bits cannot. This unique capability of qubits to exist in superpositions and entangle with each other allows quantum computers to perform certain types of calculations exponentially faster than their classical counterparts. If a classical computer were limited by memory and runtime, a quantum computer could potentially bypass these scaling issues - dubbed the "quantum advantage".

The potential of quantum computing extends beyond mere theoretical interest. If scalable quantum computers were to become a reality, they could solve some

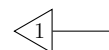
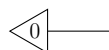
of the most challenging problems in computer science much more efficiently than classical computers. One notable example is Shor's algorithm for factoring large numbers, which could break widely-used cryptographic systems such as RSA. Another significant algorithm is Grover's search algorithm, which provides quadratic speed-ups for unstructured search problems. These advancements underscore the profound impact quantum computing could have on fields ranging from cryptography to complex optimization problems.

As we strive to advance the field of quantum computing, classical computations play an essential role in assisting and accelerating the development of quantum computers. Classical simulation is a fundamental aspect of understanding and designing quantum hardware, often serving as the only means to validate these quantum systems [1]. Additionally, quantum algorithms can be implemented on classical simulators for testing and verification purposes while full-fledged quantum computers remain beyond reach (for now). These simulators emulate the behavior of quantum computers, allowing researchers to simulate processes as if running on actual quantum devices.

This theses attempts to solve the problem in an almost entirely graphical approach. From the introduction of the qubits, to the representation of quantum circuits and linear maps, the underlying linear algebra that governs the operations of quantum computing is represented in a graphical manner. Using ZX-Calculus, a graphical calculus detailed in the next section, quantum circuits can be represented as diagrams, allowing for different discoveries to be more easily made when analysing the structure of said diagrams.

1.2 Basic Notation

In classical computing, a bit is the fundamental unit of information, and is denoted by a boolean value $b \in \{0, 1\}$. In quantum computing, the fundamental unit of information is the qubit, which we denote as $|b\rangle$, where $b \in \{0, 1\}$. Diagrammatically, they are represented as follows:



In fact, we can consider these the computational or Z-basis states.

Definition 1.2.1 (Z-basis). The Z-basis or computational basis is defined as

$$\{ \triangleleft_0 \text{---}, \triangleleft_1 \text{---} \}$$

▲

So far, the expressive power is still similar to classical computing. However, the power of quantum computing comes from the ability to exist in superpositions of these states. Concretely, the qubit can be represented in a linear combination of the basis states.

Definition 1.2.2 (Pure state). A pure state $|\psi\rangle$ is a state

$$\triangleleft_\psi \text{---} = a \triangleleft_0 \text{---} + b \triangleleft_1 \text{---}$$

where $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$.

▲

Then, the '+' or X-basis states can be defined.

$$\begin{aligned} \triangleleft_+ &= \frac{1}{\sqrt{2}} (\triangleleft_0 + \triangleleft_1) \\ \triangleleft_- &= \frac{1}{\sqrt{2}} (\triangleleft_0 - \triangleleft_1) \end{aligned}$$

Along with the Z-basis states, these states form the building blocks of understanding quantum computing and the use of ZX-calculus for diagrammatic reasoning.

Intuitively, using the diagrammatic representation, we can consider combining diagrams in parallel. In the literature, this is the same as taking a tensor product. For example, we can have a 2-qubit state.

$$\triangleleft_{10} \text{---} = \triangleleft_1 \text{---} \triangleleft_0 \text{---}$$

States can be transformed by quantum gates simply by composing them to obtain a new state. Given that composing the quantum gate U to a state $|\psi\rangle$ gives a state $|\psi'\rangle$, we have

$$\triangleleft_{\psi'} \text{---} = \triangleleft_\psi \text{---} \boxed{U} \text{---}$$

The dual of states are the effects. Here, we have the Z-basis effects.

$$\longrightarrow \triangleleft_0 \qquad \longrightarrow \triangleleft_1$$

For a state $|\psi\rangle$ and an effect $\langle\phi|$, we can obtain a scalar inner product. This is essentially the *bra-ket* notation in diagrammatic form, which comes with a few other definitions.

Definition 1.2.3 (Inner product). The *inner product* $\langle\phi|\psi\rangle$ of two states $|\psi\rangle$ and $|\phi\rangle$ is defined as

$$\triangleleft_\psi \longrightarrow \triangleleft_\phi$$

They are *orthogonal* if

$$\triangleleft_\psi \longrightarrow \triangleleft_\phi = 0$$

The *squared-norm* of a state $|\psi\rangle$ is

$$\triangleleft_\psi \longrightarrow \triangleleft_\psi$$

$|\psi\rangle$ is *normalised* if

$$\triangleleft_\psi \longrightarrow \triangleleft_\psi = \boxed{} = 1$$

where the empty diagram represents the scalar 1. ▲

We now define the inner product of the Z-basis states.

Definition 1.2.4. The inner product of the Z-basis states is

$$\triangleleft_0 \longrightarrow \triangleleft_0 = \triangleleft_1 \longrightarrow \triangleleft_1 = \boxed{}$$

$$\triangleleft_1 \longrightarrow \triangleleft_0 = \triangleleft_0 \longrightarrow \triangleleft_1 = 0$$

Succinctly, for $i, j \in \{0, 1\}$,

$$\triangleleft_i \longrightarrow \triangleleft_j = \delta_{ij}$$

where δ_{ij} is the Kronecker delta. ▲

In fact, the succinct definition can be used to determine if a basis is an *orthonormal basis* (ONB). The X-basis is also an ONB.

We are almost at the stage where we can obtain probabilities from the diagrams. The scalar inner product is not really a probability, that is, a number between 0 and 1, but instead, a number in $\mathbb{C}$. Note that putting multiple scalars in a single diagram is just the same as multiplying them together.

$$\begin{array}{cc} \triangleleft \psi & \longrightarrow \triangleleft \phi \\ \triangleleft \psi' & \longrightarrow \triangleleft \phi' \end{array} = \triangleleft \psi \longrightarrow \triangleleft \phi \cdot \triangleleft \psi' \longrightarrow \triangleleft \phi'$$

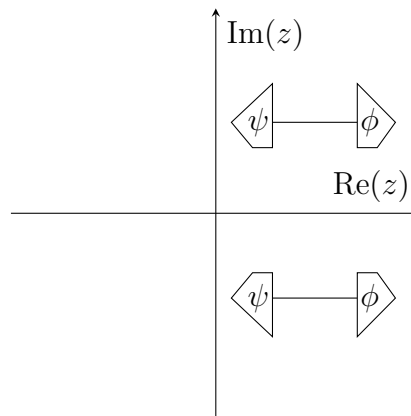
We also know that a complex number and its conjugate, $z, \bar{z} \in \mathbb{C}$,

$$|z|^2 = z\bar{z}$$

Translating that to the diagrammatic representation, we can obtain the following.

$$\left| \triangleleft \psi \longrightarrow \triangleleft \phi \right|^2 = \begin{array}{cc} \triangleleft \psi & \longrightarrow \triangleleft \phi \\ \triangleleft \psi & \longrightarrow \triangleleft \phi \end{array}$$

Note that asymmetry is introduced to denote the conjugation. Remember that a complex number $z = a + bi \in \mathbb{C}$ has a conjugate $\bar{z} = a - bi \in \mathbb{C}$, so this corresponds to a vertical reflection in the diagram above. We can use the complex plane to show why this makes sense since the reflection is about the real axis.



Using the Born rule, if we make sure that $|\psi\rangle$ and $|\phi\rangle$ are normalised, we can obtain the following.

$$0 \leq \begin{array}{c} \triangleleft \psi \\ \triangleleft \psi \end{array} \begin{array}{c} \rightarrow \phi \\ \rightarrow \phi \end{array} \leq 1$$

This can be interpreted as our desired probability - the probability of measuring $|\psi\rangle$ in the state $|\phi\rangle$.

We now have all we need to introduce the ZX-calculus notation.

1.3 ZX-Calculus

So far, we have introduced the representation of quantum states, effects, and inner products in a diagrammatic form. However, there still isn't much we can do without a way to manipulate them. In the previous section, we mentioned that we can apply quantum gates to states to obtain new states. Here, we will introduce the ZX-Calculus and show the highly intuitive representation of some quantum gates such as the Hadamard and CNOT gates, based on the building blocks from the previous section. This graphical calculus continues to use the notions of composition and tensor product from the earlier section.

1.3.1 Spiders

Definition 1.3.1 (Spiders).

$$\begin{array}{c} \vdots \\ \vdots \\ \text{---} \text{ } \alpha \text{ } \text{---} \\ \vdots \\ \vdots \end{array} := \begin{array}{c} \begin{array}{cc} \triangleleft 0 & \triangleleft 0 \end{array} \\ \begin{array}{cc} \triangleleft 0 & \triangleleft 0 \end{array} \\ \vdots \\ \begin{array}{cc} \triangleleft 0 & \triangleleft 0 \end{array} \end{array} + e^{i\alpha} \begin{array}{c} \begin{array}{cc} \triangleleft 1 & \triangleleft 1 \end{array} \\ \begin{array}{cc} \triangleleft 1 & \triangleleft 1 \end{array} \\ \vdots \\ \begin{array}{cc} \triangleleft 1 & \triangleleft 1 \end{array} \end{array}$$

$$\begin{array}{c} \vdots \\ \vdots \\ \text{---} \text{ } \alpha \text{ } \text{---} \\ \vdots \\ \vdots \end{array} := \begin{array}{c} \begin{array}{cc} \triangleleft 0 & \triangleleft 0 \end{array} \\ \begin{array}{cc} \triangleleft 0 & \triangleleft 0 \end{array} \\ \vdots \\ \begin{array}{cc} \triangleleft 0 & \triangleleft 0 \end{array} \end{array} + e^{i\alpha} \begin{array}{c} \begin{array}{cc} \triangleleft 1 & \triangleleft 1 \end{array} \\ \begin{array}{cc} \triangleleft 1 & \triangleleft 1 \end{array} \\ \vdots \\ \begin{array}{cc} \triangleleft 1 & \triangleleft 1 \end{array} \end{array}$$

▲

Here, we have green and red spiders, also known as Z and X spiders, made of the Z and X basis states respectively. The α inside a spider node is also called the

phase of the spider. For example, with $\alpha = 0$, $e^{i\alpha} = \boxed{}$. It is convenient to denote spiders with $\alpha = 0$ as having no angle written inside the spider.

Immediately, we can see that the Z and X basis states can be formed from the definition of the spiders, up to the multiplicative constant $\sqrt{2}$. Note that $e^{i\pi} = -1$.

$$\begin{aligned}
 \textcircled{0} &= \triangleleft 0 \text{---} + \triangleleft 1 \text{---} = \sqrt{2} \triangleleft 0 \text{---} \\
 \textcircled{\pi} &= \triangleleft 0 \text{---} - \triangleleft 1 \text{---} = \sqrt{2} \triangleleft 1 \text{---} \\
 \textcircled{0} &= \triangleleft 0 \text{---} + \triangleleft 1 \text{---} = \sqrt{2} \triangleleft 0 \text{---} \\
 \textcircled{\pi} &= \triangleleft 0 \text{---} - \triangleleft 1 \text{---} = \sqrt{2} \triangleleft 1 \text{---}
 \end{aligned} \tag{1.1}$$

It turns out that in reasoning with spiders, we can ignore the multiplicative constants, since much of the manipulation of the diagrams is based on the structure of the diagrams themselves. The scalars are easy to determine using the definitions we have seen so far. By using $\approx$ to denote equality up to a non-zero factor, we have

$$\begin{aligned}
 \textcircled{0} &\approx \triangleleft 0 \text{---} & \textcircled{\pi} &\approx \triangleleft 1 \text{---} \\
 \textcircled{0} &\approx \triangleleft 0 \text{---} & \textcircled{\pi} &\approx \triangleleft 1 \text{---}
 \end{aligned} \tag{1.2}$$

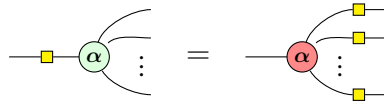
We can also see the single-qubit gates in the ZX-Calculus.

$$\begin{aligned}
 \text{---} \textcircled{\alpha} \text{---} &= \text{---} \triangleleft 0 \text{---} \triangleleft 0 \text{---} + e^{i\alpha} \text{---} \triangleleft 1 \text{---} \triangleleft 1 \text{---} = Z_\alpha \\
 \text{---} \textcircled{\alpha} \text{---} &= \text{---} \triangleleft 0 \text{---} \triangleleft 0 \text{---} + e^{i\alpha} \text{---} \triangleleft 1 \text{---} \triangleleft 1 \text{---} = X_\alpha
 \end{aligned}$$

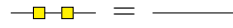
We can then define many of the common quantum gates in this way.

$$\begin{aligned}
 I &= \text{---} \textcircled{0} \text{---} = \text{---} \textcircled{\pi} \text{---} = \text{---} \\
 X &= \text{---} \textcircled{\pi} \text{---} \\
 Z &= \text{---} \textcircled{0} \text{---} \\
 H &= e^{-i\frac{\pi}{4}} \textcircled{\frac{\pi}{2}} \textcircled{\frac{\pi}{2}} \textcircled{\frac{\pi}{2}} \textcircled{\frac{\pi}{2}} =: \text{---} \textcircled{} \text{---} \equiv \text{---} \text{---} \text{---} \\
 S &= \text{---} \textcircled{\frac{\pi}{2}} \text{---} \\
 T &= \text{---} \textcircled{\frac{\pi}{4}} \text{---} \\
 CNOT &= \text{---} \textcircled{0} \text{---} \\
 &\quad \text{---} \textcircled{\pi} \text{---}
 \end{aligned} \tag{1.3}$$

There are 2 gates of special note here. The Hadamard gate H , denoted by the small yellow box, or as a blue dashed line, is a single-qubit gate that interchanges Z and X bases, so it acts as a colour changer.



It is also a self-inverse.



The CNOT gate is a two-qubit gate that acts on the second qubit if the first qubit is in the state $|1\rangle$. We will see when introducing the cups and caps, what the vertical line in the CNOT gate represents.

*Alles Gescheite ist schon gedacht worden.
 Man muss nur versuchen, es noch einmal zu denken.*

*All intelligent thoughts have already been thought;
 what is necessary is only to try to think them again.*

— Johann Wolfgang von Goethe [2]

2

Background

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2.1 Introduction

The ZX-Calculus is a graphical calculus for quantum computations, developed by Coecke and Duncan in 2008 [3]. All operations are expressed as *spiders*, connected by *edges* (or wires).

This document introduction won't serve as a complete primer on L^AT_EX. There are plenty of those online, and googling your questions will often get you answers, especially from <http://tex.stackexchange.com>.

Instead, let's talk a little about a few of the features and packages lumped into this template situation. The `savequote` environment at the beginning of chapters can add some wittiness to your thesis. If you don't like the quotes, just remove that block.

For when it comes time to do corrections, there are two useful commands

here. First, the `mccorrect` command allows you to highlight a short correction like this one. When the thesis is typeset normally, the correction will just appear as part of the text. However, when you declare `\correctionstrue` in the main `Oxford_Thesis.tex` file, that correction will be highlighted in blue. That might be useful for submitting a post-viva, corrected copy to your examiners so they can quickly verify you've completed the task.

For larger chunks, like this paragraph or indeed entire figures, you can use the `mccorrection` environment. This environment highlights paragraph-sized and larger blocks with the same blue colour.

Read through the `Oxford_Thesis.tex` file to see the various options for one- and two-sided printing, including or excluding the separate abstract page, and turning corrections and draft footer on or off, and the separate option to centre your text on the page (for PDF submission) or offset it (for binding). There is also a separate option for master's degree submissions, which changes identifying information to candidate number and includes a word count. (Unfortunately, $\text{\LaTeX}$ has a hard time doing word counts automatically, so you'll have to enter the count manually if you require this.)

2.2 Cardiac Imaging

Within months of Röntgen's discovery of the X-ray in 1895[4], cardiac pathology was being investigated via non-invasive imaging [5]. Over the intervening years, cardiac imaging modalities and techniques have advanced significantly. Clinically, cardiac imaging is used for two broad purposes: diagnosis of pathophysiology and guidance of interventional procedures. These applications impose different requirements on imaging equipment, image acquisition time, computational complexity, spatial and temporal resolution, and tissue discrimination. The common diagnostic and interventional cardiac imaging techniques in current clinical practice are reviewed below. An accessible introduction to the physics of medical imaging can be found in Webb's *Introduction to Biomedical Imaging* [6]. A comprehensive overview of the use of imaging in clinical cardiology is presented in Leeson's *Cardiovascular Imaging* [7].

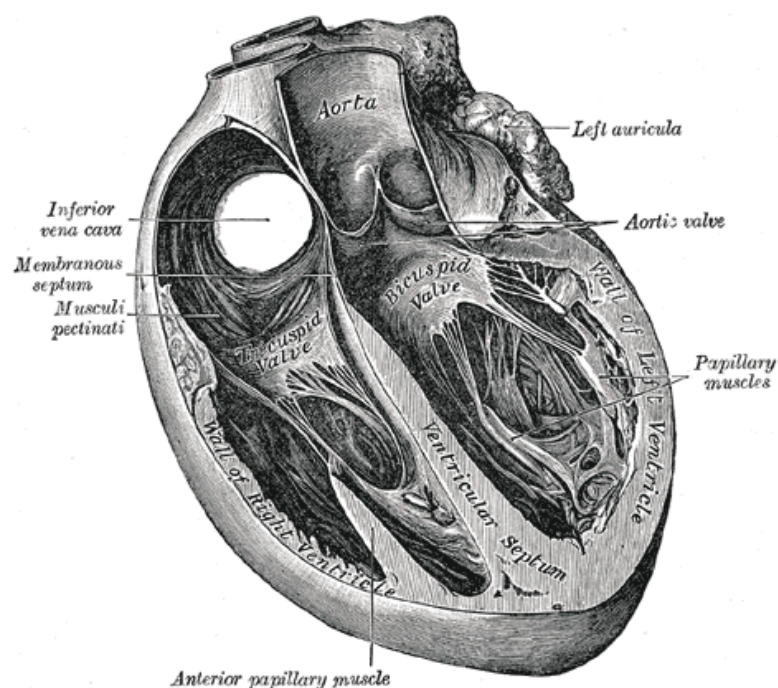


Figure 2.1: Four-chamber illustration of the human heart. Clockwise from upper-left: right atrium, left atrium, left ventricle, right ventricle.

2.2.1 Diagnostic Imaging

Beyond the chest X-ray (‘plain film’), the key non-invasive imaging modalities in diagnostic cardiology are echocardiography, magnetic resonance imaging, and X-ray computed tomography, which are reviewed below. Nuclear medicine, including positron emission tomography (PET) and single-photon emission computed tomography (SPECT), are not discussed here, as they do not play a role in the chapters to follow.

Echocardiography

The use of acoustic waves for medical diagnosis, inspired by naval sonar, was initially developed in the 1940s [8]. By 1954, the first clinically useful cardiac ultrasound – examining motion of the mitral valve in stenosis – was reported [9]. These early scans were one-dimensional images (‘A-mode’), sometimes repeated to generate a time axis (‘M-mode’). The sector-scanning probe was developed in the 1970s [10, 11], leading to the ‘B-mode’ that a modern cardiologist would recognise as an echocardiogram.

Appendices

Cor animalium, fundamentum est vitæ, princeps omnium, Microcofimi Sol, a quo omnis vegetatio dependet, vigor omnis & robur emanat.

The heart of animals is the foundation of their life, the sovereign of everything within them, the sun of their microcosm, that upon which all growth depends, from which all power proceeds.

— William Harvey [12]



Review of Cardiac Physiology and Electrophysiology

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Appendices are just like chapters. Their sections and subsections get numbered and included in the table of contents; figures and equations and tables added up, etc. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed et dui sem. Aliquam dictum et ante ut semper. Donec sollicitudin sed quam at aliquet. Sed maximus diam elementum justo auctor, eget volutpat elit eleifend. Curabitur hendrerit ligula in erat feugiat, at rutrum risus suscipit. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Integer risus nulla, facilisis eget lacinia a, pretium mattis metus. Vestibulum aliquam varius ligula nec consectetur. Maecenas ac ipsum odio. Cras ac elit consequat, eleifend ipsum sodales, euismod nunc. Nam vitae tempor enim, sit amet eleifend nisi. Etiam at erat vel neque consequat.

A.1 Anatomy

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A.2 Mechanical Cycle

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A.3 Electrical Cycle

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A.4 Cellular Electromechanical Coupling

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*The first kind of intellectual and artistic personality
belongs to the hedgehogs, the second to the foxes ...*

— Sir Isaiah Berlin [13]

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